

Exercise 3

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Proof. 1. $\implies$: Suppose $\forall g \in S_k(1)$, $a_g(n) = O(n^{(k-1)/2+\epsilon})$. For Hecke Eigenforms $f \in H_k \subset S_k(1)$, $a_f(n) = \lambda_f(n)$. By b, $\lambda_f(n) \leq d(n)n^{(k-1)/2+\epsilon}$.

2. $\impliedby$: H_k is a eigenbasis of $S_k(1)$ a finite vectorspace. Thus $\{f_1, \dots, f_{\dim S_k}\} = H_k$ and $\forall g \in S_k(1)$

$$g = c_1 f_1 + \dots + c_{\dim S_k} f_{\dim S_k}$$

$$\text{Thus } a_g(n) = \sum_{j=1}^{\dim S_k} c_j a_{f_j}(n) = \sum_{j=1}^{\dim S_k} c_j \lambda_{f_j}(n) \leq d(n)n^{(k-1)/2} \sum_{j=1}^{\dim S_k} c_j$$

□